

# Yohei Ohata Curriculum Vitae

Center for Information and Neural Networks  
National Institute of Information and Communications Technology  
1-4, Yamadaoka, Suita, Osaka, Japan

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## Education

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### The University of Tokyo

April 2022 – March 2024

M.S. in Human Factors Engineering

Department of Human and Engineered Environmental Studies

Cumulative GPA: 4.0/4.0

*Thesis:* Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models

1st in Class (1/48), Dean's Award, Best Department Thesis Award

### Tokyo University of Science

April 2018 – March 2022

B.E. in Electrical and Electronic Engineering

Department of Applied Electronics

Cumulative GPA: 3.5/4.0

*Thesis:* A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions

## Employment

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### National Institute of Information and Communications Technology

January 2026 – Present

Research Scientist

Center for Information and Neural Networks (CiNet)

Computational Social Neuroscience Group

### Hitachi, Ltd.,

April 2024 – December 2025

Research Scientist

R&D Group

## Publications

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### Articles in Peer-Reviewed Journals

- J1 Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (submitted). Identifying Different Thought Processes for Each Learner Using Eye-Tracking and Hidden Markov Models. *Human Factors*.

### Peer-Reviewed Conference Proceedings

- C1 Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2024). Visualizing and Revealing the Difference of Learner's Thought Process and Question Solving Method. *In Proceedings of International Conference on Information and Communication Technology*.
- C2 Ohata, Y.**, Takao, K., Natori, T., & Aikawa, N (2021). A Method of Designing a Hilbert Transformer with Transmission Zeros at Specified Positions. *ICICE Technical Report*.

## Conference Presentations

- P1 Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2023). Analysis of How Learners Learn Using Graphic Questions. *The 42nd Annual Conference of Japanese Society for Educational Technology*.
- P2 Ohata, Y.**, Hachisuka, S., Kurita, K., & Warisawa, S. (2022). Identifying Perceptual Factors that Influence Impression of a Teacher in a Classroom. *The 41st Annual Conference of Japanese Society for Educational Technology*.

## Patents

- T1 Ohata, Y.**, Jin, Y. (2024). “Management Issues Estimation Device”. JP-Patent.
- T2 Ohata, Y.**, Jin, Y. (2025). “Management Issues Estimation System Using Bayesian Network and Large Language Models”. JP-Patent.

## Research Experience

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### Research Scientist

January 2026 – Present

National Institute of Information and Communications Technology  
Laboratory: Computational Neuroscience & Decision Making Laboratory  
*Supervisor: Haruno Masahiko*

- Investigated brain decoding foundation models incorporating attention mechanisms and VQ-VAE, while exploring latent variable frameworks such as CEBRA and FINDR for neural data analysis

### Research Assistant

September 2025 – Present

The University of Tokyo  
Laboratory: Amano & Nakayama Lab  
*Supervisor: Yamashita Ayumu, Amano Kaoru*

- Performed denoising on the DMCC55B Open fMRI dataset (AX-CPT, Cued Task Switching, Sternberg Working Memory, Stroop) following the Ciric et al. (2017) pipeline and synchronized cross-subject temporal variability using BrainSync (Akrami et al. (2019)) (MATLAB).
- Applied NASCAR tensor decomposition to the DMCC55B dataset to decompose data into spatial, temporal, and subject-load modes (MATLAB).
- Quantified performance by indexing RT and Accuracy via the Cognitive Control Index (CCI), extracting contributory components using dCRT with d0 statistics (R).
- Evaluated spatial map similarity across cognitive tasks using t-SNE, ISOMAP, UMAP and feature selection via Nested Cross-Validation LASSO (Python).

### Research Assistant

April 2022 – March 2024

The University of Tokyo  
Laboratory: Inovative Learning Creation Lab  
*Supervisor: Hachisuka Satori, Warisawa Shin'ichi*

- Designed experimental protocols for both adult and elementary school participants, including the development of a web-based UI and logging system using HTML/CSS/JS.
- Managed high-fidelity eye-tracking data collection and processing using Tobii Pro 3 and Tobii Pro Lab.
- Performed advanced statistical analysis using Linear Mixed-Effects Models (LMM) in R.
- Modeled the temporal dynamics of scan paths using Hidden Markov Models (HMM) implemented in Python.
- Conceptualized research goals, authored manuscripts.

## Research Assistant

Tokyo University of Science  
Laboratory: Signal Processing Lab  
*Supervisor: Aikawa Naoyuki*

May 2021 – December 2021

- Proposed a Composite Hilbert Transformer that integrates a pre-normalized notch filter to suppress noise at specific frequencies.
- Verified significant SNR improvements over conventional Band Pass Hilbert Transformer through extensive simulations in MATLAB.

## Professional Experience

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### Research Scientist

Hitachi, Ltd., Digital Innovation R&D

April 2024 – December 2025

- Developed concepts and business use cases
- Defined functional requirements and methodologies
- Authored patent applications
- Applied LLMs in service development
- Established evaluation criteria for LLM verification

## Teaching Experience

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### Sensing and Signal Processing

Teaching Assistant, The University of Tokyo

April 2023 – June 2023

- Assisted in building signal processing systems using LabVIEW

## Honors and Awards

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IEEE CAS JJC Best Student Award

2021

## Languages

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- High Level in English
- Native in Japanese

## References

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### Hachisuka Satori, Ph.D.,

Associate Professor, The University of Tokyo  
Email: hachisuka@edu.k.u-tokyo

### Yamashita Ayumu, Ph.D.,

Assistant Professor, Advanced Telecommunications Research Institute International  
Email: ayumu@atr.jp

### Haruno Masahiko, Ph.D.,

Professor, National Institute of Information and Communications Technology  
Email: mharuno@nict.go.jp

